Tigo Deken

Engineering Product Design, HvA Amsterdam. Looking for a 5-month internship in NPI, manufacturing or injection moulding DFM from February 2027.

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Project

Recharge Flip Cap

Course Verpakkingsontwerp (packaging design), HvA	Year
My role Individual	Material PP shell, HDPE refill
Process Moulded packaging, SolidWorks CAD	Status Completed

A refillable shampoo bottle where the PP outer shell stays and only a light HDPE cartridge is replaced, rated "good recyclable" in the recycle check.

Manufacturing skill: **design for recycling, packaging DFM**

Problem and constraints

Marie Stella Maris sells shampoo in premium bottles. Once empty, the whole bottle is thrown away. The brief asked for a refill system that keeps the premium feel and cuts plastic waste.

The brief set hard material rules:

- Mono-material parts: no labels, glue, metal parts, silicones or elastomers.
- Base and pump reusable several times; only the refill cup is replaced.
- The bottle has to stand stably on a bathroom shelf.

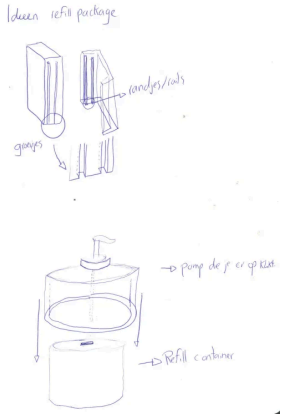
My role

I did the concept sketches, the SolidWorks model, the technical drawings and the sustainability check.

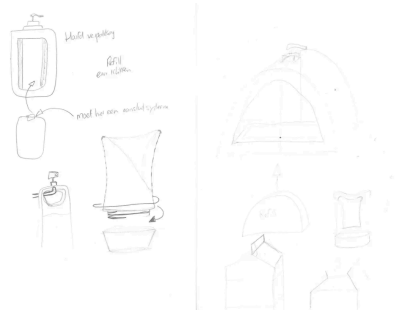
Method

I explored refill mechanics and connection systems on paper first, then settled on a click-in refill cartridge inside a domed outer shell.

- **Two parts, two jobs.** The outer shell in PP is the part the user keeps. The refill cartridge in HDPE is the part that travels through the supply chain.
- **No mixed materials.** Branding sits on the bottom as moulded-in text, so no label or glue is needed.
- **Stability.** A widened base and low centre of gravity keep the bottle upright.
- **Check.** I ran the design through a recycle check, a circularity check and an environmental impact check.



Early sketches of refill packages and connection ideas.



Refined concepts, moving towards a domed outer shell.

Results

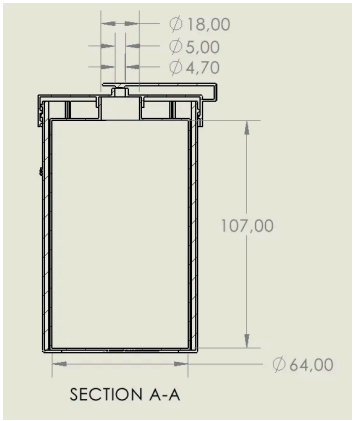
- Body Ø64 mm, height 107 mm (section A-A).
- Refill cartridge Ø66 mm by 109 mm on the multi-view drawing.
- Recycle check: "good recyclable".
- Fee modulation indication: €0.30 in total, €0.10 each for mono-material PP/PE/PET, a minimum share of post-consumer recycled plastic, and the "good recyclable" score.
- Around 80% less plastic, by keeping the shell and replacing only the cartridge.



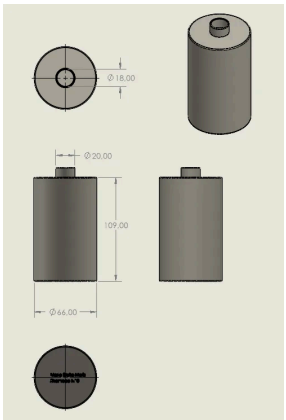
SolidWorks model, front view.



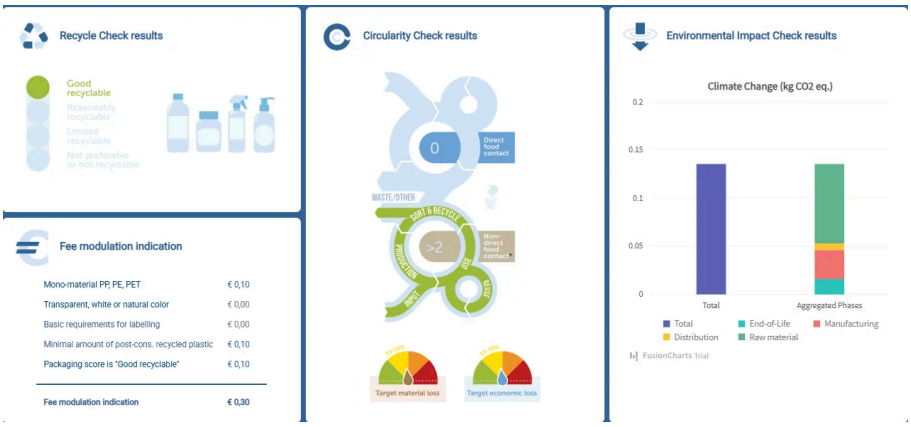
Flip cap detail.



Section A-A: body Ø64 mm, height 107 mm, cap features Ø18, Ø5 and Ø4.70 mm.



Multi-view drawing of the cartridge with main dimensions.



Recycle check, fee modulation, circularity check and climate impact by life-cycle phase.

What I would change

- Shell and cartridge are two different polymers (PP and HDPE). Each part is mono-material, but the user has to separate them for recycling.
- The cartridge-to-shell fit needs a tolerance study: too tight and the refill is hard to swap, too loose and it rattles.
- Test a printed prototype for stability and the flip cap feel, instead of judging them from the render.

Files

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- CAD renders and drawings: shown above.

Project

Air intake for carburetted motorcycles

Type Personal project	Year 2026
My role Individual	Material Prototype: 3D printed. Final: under evaluation
Process FDM prototyping, Fusion 360	Status In progress

A replacement intake for older carburetted bikes that bolts onto the existing carburettor and boot. Concept stage: no test results yet.

Manufacturing skill: **fit-up tolerances, prototype testing**

Problem and constraints

Older carburetted motorcycles often run with a cracked or missing original airbox, or with a universal pod filter. A pod filter fits easily but changes how air reaches the carburettor, which affects the mixture. I want an intake that fits a specific bike and keeps the carburettor working as intended.

- Fit the existing carburettor mouth and intake boot without modifying the bike.
- Survive engine heat, vibration, fuel vapour and rain.
- Keep the filter serviceable with basic tools.
- Be reversible, so the bike can go back to standard.

My role

Personal project, done on my own: research, CAD, prototyping and testing.

Method

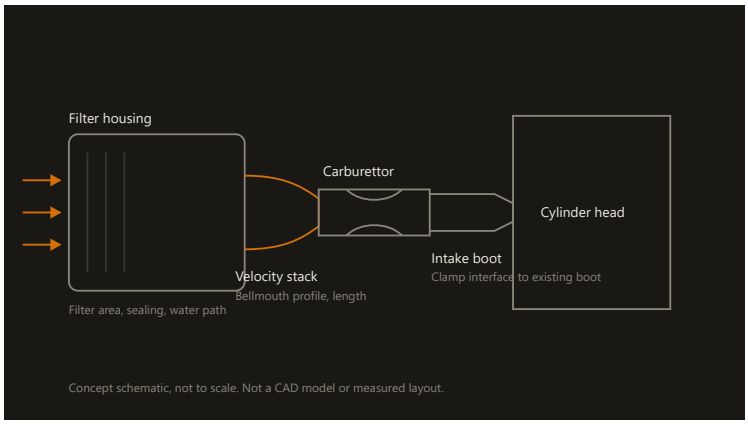
The intake breaks down into three parts, each with its own design question:

- **Carburettor interface.** Measure the carburettor mouth and boot, then design a clamped sleeve with a fit that seals without forcing the rubber.
- **Velocity stack.** A bellmouth entry to guide air into the carburettor. Its profile and length are the main variables.
- **Filter housing.** Sized around an off-the-shelf filter element, with a drain path for water and a cover that comes off without removing the intake.

Material for the final part depends on the temperatures measured near the cylinder. A printed part is only a fit and flow prototype.

Planned tests:

- Fit check of the printed interface on the bike.
- Surface temperature near the intake with the engine warm.
- Back-to-back ride comparison against the current set-up.



Concept schematic of the intake path and the design question for each part. Not to scale.

Results

What I would change

Files

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Project

Plastic motorcycle body parts

Type Personal project	Year 2026
My role Individual	Material ABS, PC/ABS, PP compared
Process Injection moulding (design study), Fusion 360 simulation	Status In progress

A material comparison and moulding design study for plastic side covers and fairing parts. Analysis stage: no test results yet.

Manufacturing skill: **material selection, injection moulding DFM**

Problem and constraints

Side covers and fairing parts on motorcycles are injection-moulded plastic. They crack at mounting points, fade in the sun and are often no longer sold for older bikes. I want to understand why they fail and how I would design and specify a replacement.

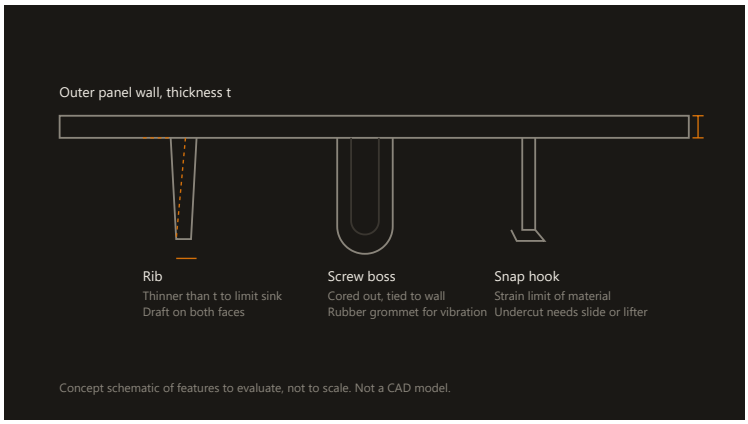
- Impact resistance, also in cold weather.
- UV and weather resistance, and a surface that can be painted.
- Heat near the engine and exhaust.
- Mounting points that survive vibration.
- A shape that can be moulded: even walls, draft, no needless undercuts.

My role

Personal project, done on my own: material research, CAD, simulation and prototyping.

Method

- **Failure review.** Look at where existing parts crack and fade, and link each failure to a load or exposure.
- **Material selection.** Compare ABS, PC/ABS and PP on impact strength, heat deflection temperature, UV resistance, paintability and cost, using datasheet values.
- **Moulding design.** Redesign the part for injection moulding: constant wall thickness, draft on all faces in the pull direction, ribs thinner than the wall to avoid sink marks, cored screw bosses.
- **Mounting.** Compare a screw boss with a rubber grommet against a snap hook, and check the snap hook strain against the material limit.
- **Simulation.** Static stress on a mounting tab in Fusion 360, then a printed prototype for fit.



Concept schematic of the moulded features to evaluate on the panel. Not to scale.

Results

What I would change

Files

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